Chapter 16, Problem 92P

Problem

Develop the state equations for the following differential equation.

$$\frac{d^2y(t)}{dt^2} + \frac{7\ dy(t)}{dt} + 9y(t) = \frac{dz(t)}{dt} + z(t)$$

Step-by-step solution

Step 1 of 1

$$Let \quad X_1 = y(t) \quad And$$

$$X_2 = X_1' - 2 = Y' - Z \quad or \quad Y' = X_2 + Z$$

$$Thus \quad X_2' = Y'' - Z' = -6X_1 - 5(X_2 + Z) + Z' + 2Z - Z'$$

$$= -6X_1 - 5X_2 - 3Z$$

This now leads to our state equations;

$$\begin{bmatrix} X_1' \\ X_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \end{bmatrix} Z(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + [0]Z(t)$$